



some other manner. This is done by including a header along with each packet that contains the information on the file it represents, including its length. This also tells the OS where the next packet begins. In this way, the OS can read the CD contents. However, adding this additional information decreases the disc's capacity.

**Cross-talk:** A measure of the amount of interference coming from neighbouring pit tracks on a CD. As track pitch is tightened (when tracks are packed closer together to put more data on a disc), cross-talk increases. A maximum value of 50 per cent is allowed by Red Book specifications.

**Mastering:** The process of creating a stamper or a set of stampers to be used in the injection moulding stage of manufacturing compact discs. A digital signal from a PC is used to guide a laser beam, which etches a pattern of 'pits and lands' (in the case of CDs) or a continuous groove (for CD-Rs) onto a highly polished glass disc coated with photo-resist.

This 'glass master' is then cured (developed) with ultraviolet light and rinsed off, and a metal (nickel or silver) mould is electroformed on top of it. This mould is removed and then electroplated with a nickel alloy to create one or more stampers to be used in the injection moulding machine to press the data into

the polycarbonate substrate of CDs, or the guiding groove into the CD-R substrate.

**Media or 'blanks':** CD-R media are the discs used to record digital information with a special recorder and pre-mastering software on a computer. These discs are made of a polycarbonate substrate, a layer of organic dye, a metalised reflective layer, and a protective lacquer coating.

Some discs also have an additional protective coating over the metalised layer, while others have a printable surface silk screened on them.

**Organic dye:** The data layer of CD-R discs is made from either cyanine or phthalocyanine dye, which is melted during the recording process. It then becomes opaque or refractive, scattering the reading laser beam so that it does not reflect into the reader's sensors. The difference between reflected and non-reflected light is interpreted by the player as a binary signal.

**Pits and lands:** In a 'pressed' or mass-replicated CD, the bumps and grooves that represent the binary data on a disc's substrate are pressed into it during manufacture. CD-R discs do not have true pits and lands, but the un-melted, clear areas and melted, opaque places in the dye layer fulfill the same function as pits and lands on a pressed disc.

## CD-ROM File System



To access any file, a computer's operating system needs to be able to locate the file. A file system facilitates the storage and retrieval of hundreds or even thousands of distinct files. Every operating system uses a different file system and therefore, there are a number of CD-ROM file systems in use on Windows, Macintosh, Unix and others. The most common file systems for CD-ROMs are the ISO 9660 and Joliet extensions. ISO 9660 is the international standard and is

designed for the PC and MS-DOS. The Joliet specification was designed to resolve certain deficiencies of the original ISO 9660 file system, such as a limited character set and file name and path lengths. CD-ROMs that are to be used on Macintosh computers are usually created in the Hierarchical Filing System (HFS) format. This format features support for sub-directories, up to 31 characters per file name and up to 27 characters for volume names.

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